

Development of an Intravenous Fluid Monitoring, Warning, and Reverse Flow Blocking System

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Abstract— With the advancement of medical technology, the improvement of healthcare requires the integration of cutting-edge technology with medical science to make an effective system. Medical care has increased significantly in recent years as a consequence of technological breakthroughs. A fundamental demand for any hospitalized patient includes advanced care and continuous monitoring. Intravenous (IV) treatment is the fastest and most common way of treatment since it skips the digestive system. When saline is supplied to a patient, the patient is administered it continuously by a nurse. Unfortunately, the nurse could forget to turn off the saline injection system for a variety of reasons, including hectic schedules, forgetfulness, or sleeping at night. The saline tubing system is then re-flexed by the blood of the patient, endangering their lives. Additionally, air bubbles can enter the patient's body. Consequently, there is a necessity for an automated system for assuring the continual monitoring and regulation of the fluid flow. To answer this, we have developed a remotely monitored saline level and automated alarm system to keep patients safe and preserve their lives during saline feeding hours by employing a microcontroller, load cell, switching valve, and GSM module. The proposed system has a weight measuring sensor that monitors the weight of the saline bottle and processes that weight via a microcontroller unit, which subsequently signals when the saline bottle is empty and automatically shuts off the saline flow when it is completely empty. This system is highly beneficial for accurate intravenous treatment, especially at night.

Index Terms— Intravenous fluid monitoring, Alarm system, Reverse flow blocking, GSM module.

I. INTRODUCTION

According to recent Worldwide Health Observatory (WHO) data on physician density per population, the global ratio in the Asian Subcontinent area is less than one physician for every 1000 people [1]. In many nations, the present nurse shortage situation is expected to quickly deteriorate in the upcoming years. Because of population growth, traditional health-care practices are becoming outdated. Researchers are trying to innovate solutions to problems with available and newer technologies at a lower cost. To make the healthcare system smart, the processes of diagnosis, treatment, management, and decision-making must be automated as much as possible so that services are provided to both rural and urban residents [2]. Due to a scarcity of care, professionals are struggling to continuously monitor and

check on the large patient population. A caretaker or nurse is almost always responsible for keeping a check on the saline bag level, and if they do not monitor this, the patient will suffer. When the fluid in the drip bag is about to run out, the nurses are usually only made aware of the situation by visiting the wards on a regular basis to check on the patient or by a request call from the patient's companion beside the bed. Such a technique has resulted in frequent interruptions of nurses' activities as well as a rise in potential costs. When a saline bottle is empty and the needle is not withdrawn from the patient's vein, blood flows outward into the bottle owing to pressure differences, which can result in significant injury. As a result, it is necessary to automate surveillance in order to avoid such a hazard. As a result, there has recently been interest in the development of many feasible prototypes for monitoring automatic intravenous drip closure. Furthermore, long-distance monitoring by a doctor is required for telehealth services [3].

Some researchers developed an alert system using IoT-based technology and got some good results. In [4] and [5], the authors proposed an IoT-based IV fluid infusion rate monitoring system where IV fluid is detected by an IR sensor and flow is stopped by using some mechanical design with a servo motor. But the IR setup used by their project to measure the fluid level is very sensitive, and if the reference level of the IR sensor changes for any external reason, then it will become useless for droplet counting and give an inaccurate result. Obviously, this IoT-based web monitoring and controlling setup makes life easier and more comfortable, but there is some uncertainty about Wi-Fi connectivity while monitoring [2] [6]. If Wi-Fi connectivity is disrupted, this monitoring system will be useless in a critical situation. As a result of an internet connection problem or server problems, if a fluid bag is emptied and it's not notified to the medical assistant, then it may threaten patient life. S. Velmurugan et al. developed an automated system of fluid control with a container changing system. Where multiple containers are connected in parallel and the fluid level is monitored, when one bottle runs out of fluid, another bottle begins to flow. Similarly, the entire system is dependent on internet connectivity, so any disruption in cloud connectivity will result in a critical issue [3]. M. K. Bhavasaar et al. proposed a wireless protocol design with transceiver modules for

Zigbee that make personal area networks and communicate between modules through channels that work without any interference [7]. Huang et al. developed a concept based on RFID technology to provide an automatic warning system when the fluid is running low. They proposed a personalized RFID tag that is injected into the fluid bag when it is full or nearly empty. The tag is disabled to respond to the remote RFID reader. However, when the fluid level within the bag is below the tag, the tag operates normally again and echoes to the reader to communicate an empty status [8]. As the tag antenna is radiation-dependent, it may be affected by high-density materials like metal bottles, juice liquids, etc. A health monitoring system has been used to detect the threshold level of fluid in [9]. After crossing the critical level, it notifies the assistant through SMS using the GSM module. Similar patient monitoring projects were proposed by Joseph [10] and Mansour [11], in which data was transmitted via the Bluetooth module. But unfortunately, Bluetooth's range is about 10 meters without any obstacles, so it will not be appropriate at all in noisy places like medical wards. K. R. Rani et al. used the NRF24L01 transceiver module instead of Bluetooth communication to transmit data over a distance of nearly 800 meters within the local network [12].

Automation makes healthcare administration more sustainable, scalable, and viable as the population grows. This study implements a novel technical approach to the monitoring and reporting of intravenous fluid or saline fluid flow. With the use of Global System for Mobile Communication (GSM)-based technology, our proposed work introduces a portable system that is capable of measuring the weight of a fluid container by load cell technique and monitoring the flow rate. After crossing a threshold level, it automatically notifies the assistant about the fluid level of the container and about changing the container with a specific bed number through mobile SMS, so wherever the assistant is, he/she will be notified as well as alerted locally by an LED light and buzzer. Afterward, if the container is not changed after a certain period, the valve will automatically close and prevent reverse IV fluid flow. As a result, there is no doubt about connectivity because GSM technology is more reliable than any other form of cloud connectivity. So basically, this proposed system overcomes previous drawbacks and creates a user-friendly interface for everyone. Furthermore, it can be used commercially. This user-friendly interface not only overcomes previous drawbacks but also is designed to help medical assistants, who always have a busy schedule and serve numerous patients in the wards while performing various multitasking tasks.

The rest of the manuscript is arranged as follows: The main concept is depicted as a block diagram in Section II; most essential accessories of the reverse flow blocking system have been introduced in Section III; and Section IV describes the prototype of the system. A conclusion has been drawn in Section V.

II. CONCEPT MAKING AND SYSTEM MODELLING

Today, automation is deeply intertwined with humans, including automated cars, manufacturing, automated car parking, IoT homes, automated irrigation systems, and many other routine tasks controlled by technology, but healthcare has not been touched by automation. Many services are still delivered manually in every healthcare setting. Telehealth care is playing a special role in creating better healthcare today, and it needs to be made suitable for automation in the future as well [3]. Every day, new research is conducted in order to solve new problems or update the system. This research idea was inspired by personal experience. Our aunt recently gave birth to a healthy baby in the month of Ramadan. She had a successful C-section procedure. To aid in her recovery following the procedure, the doctor prescribed IV therapy. The idea was to mix medication with saline, which would feed the body minerals. We had to keep a close eye on the IV fluid bottle. Because it was Ramadan, monitoring the IV bottle at night was really difficult. The venous blood once reflexed back to the bottle after the saline was emptied because the venous blood pressure was higher than the bottle pressure. For us, this phenomenon was terrifying. We called the nurse, but due to a staff shortage at the hospital, the nurse assigned to look after our patient arrived much later than expected. A lot of blood was lost by the patient. The patient experienced trauma from this occurrence. We created and tested a prototype device to watch, warn against, and stop the reverse blood flow from the vein in order to prevent this scenario from ever occurring. This work has been proposed after facing this problem and finding its solution.

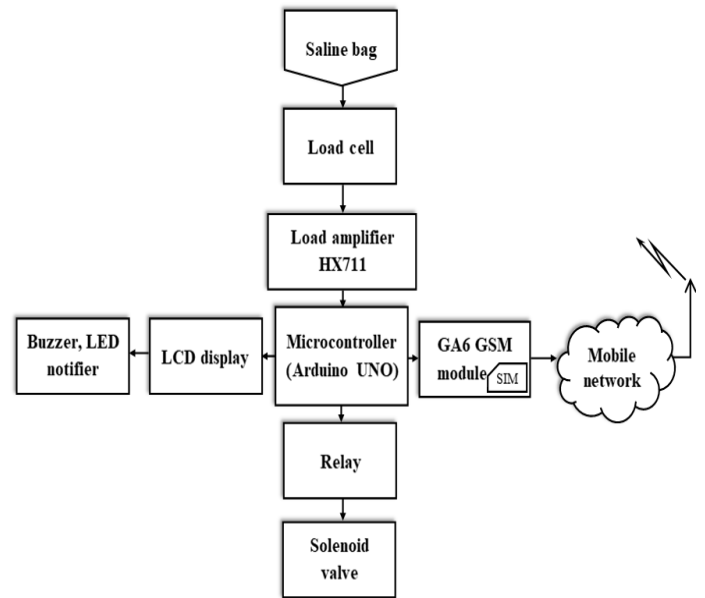


Fig. 1. Block diagram of the proposed monitoring system

A block diagram of the intravenous fluid monitoring and reverse blood flow blocking system is depicted in Fig. 1. This system encourages a GSM-based automatic alerting and reverse blood flow blocking system, in which a load cell sensor works to measure the weight of the saline bag (IV fluid bag). When the saline level falls, the weight of the saline bag

decreases as well. The load sensor's output continuously delivers the response of weight to the microcontroller (Arduino), and the display shows the percentage of liquid in the bottle used up. The microcontroller determines if the fluid level is too low compared to the input data from the load sensor and the predetermined threshold value (15% of the total bag weight) from the microcontroller. Then, locally, it responds with a buzzer sound and an LED blink notification. It can be used to alert a patient relative who is looking after the patient when 10% of the total bag weight comes around, and then a message is sent with a phone call to notify the nurse that the saline supplied to the patient has run out via a GSM module (GA6-B). If the nurse does not quickly respond to the patient and the saline level comes to 7%, then a line blockage arrangement is made automatically utilizing a solenoid valve. By shutting the valves, the saline tube is blocked. This concept stops saline and blood from flowing upward from the veins to the bottle.

III. ACCESSORIES OF THE PROPOSED SYSTEM

A specified number of electrical components must be employed to achieve a specific goal. The components are required to be correctly connected in order to operate synchronously. In this section, the major components like the Arduino Uno, the weight sensor, the LCD module, the solenoid valve, and the GSM module of the designed prototype have been briefly discussed.

3.1 Arduino Uno

The Arduino Uno is a user-friendly microcontroller board based on the ATmega328P IC, which serves as the system's brain. It is an 8-bit microcontroller operating at 16 MHz and consists of 14 GPIO pins. Use I2C, SPI, and GPIO pins to communicate with other devices. The Serial Peripheral Interface (SPI) is a synchronous serial communication interface specification used for short-distance communication, primarily in embedded systems. This unit is integrated with 1KB EEPROM and 2KB SRAM. All sensor units in this work are powered by a 5V regulated external power supply. The Arduino board has a programming environment that supports the C or C++ programming languages [13] [14].

3.2 Liquid Crystal Display (LCD)

The function of the LCD is to get the data from the microcontroller and display the information as programmed in the microcontroller. A 16×2 LCD has 16 columns and 2 rows, so it can display 32 characters (16×2 = 32) in total, and every character will be made with 5×8 (40) pixel dots. It has 16 pins that can work in both 8-bit and 4-bit modes. But by additionally attaching an I2C module to the 16×2 LCD, you can save a lot of pins since it uses only 4 pins to connect to the microcontroller. This module features a PCF8574 chip (for I2C communication) and a potentiometer to adjust the LED backlight. It works as an intermediary between the LCD and an MCU (Arduino). Every device that can be attached to an MCU has an address and can communicate with that particular device. An I2C LCD has only 4 pins that interface it to the outside world. The connections are as follows [15]

[16]: GND is a ground pin that should be linked to the Arduino's ground. V_{cc} supplies power to the module and the LCD. Connect it to the 5V output of the Arduino or a separate power supply. SDA is a serial data pin. This line is used for both transmitting and receiving. Connect to the SDA pin on the Arduino. SCL is a serial clock pin. This is a timing signal supplied by the bus master device, and it connects to the SCL pin on the Arduino.

3.3 GA6-B

The GA6 module, which is based on the GPRS A6 chip, is a scaled-down version of the serial GSM/GPRS core development board. The GA6-B has quad band frequencies of 850, 900, 1800, and 1900 MHz. The frequency bands can be set by AT commands. The GSM/GPRS modem includes an internal TCP/IP stack that allows it to easily connect to the internet via GPRS. GSM stands for "Global System for Mobile Communications" and is the global standard for mobile communications. GPRS stands for "General Packet Radio Service". GPRS is a mobile service on 2G and 3G cellular networks. The GA6-B is an ultra-compact and reliable wireless module. Communication with this module is done through the UART (Tx, Rx) interface [17].

3.4 Weight Sensor

The load cell is employed in this study as a weight measurement sensor. A load cell converts a force into a measurable electrical output. The electrical signal varies in response to the applied force. This response is routed to the HX711 load cell amplifier via a dual-channel, 24-bit precision analogue-to-digital weight pressure amplifier. The cables leaving from the load cell are directly connected to the HX711. The HX711 uses a two-wire interface to communicate with the microcontroller (clock and data). Libraries have been used to simplify reading data from the HX711 and should work with any microcontroller's GPIO pins. The connection diagram from the load cell to the Arduino board [18] is presented in Fig. 2 as follows:

Load Cell	→	HX711	→	HX711	→	Arduino
Red (E+)	→	E+	→	GND	→	GND
Black (E-)	→	E-	→	DT	→	Pin 2
White (A-)	→	A-	→	SCK	→	Pin 3
Green (A+)	→	A+	→	V _{CC}	→	5V

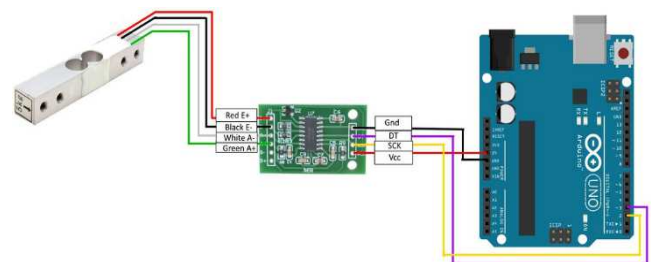


Fig. 2. Connection diagram from the load cell to the Arduino board

3.5 Solenoid Valve

Here, a 12V DC solenoid valve is employed to regulate liquid flow and serves as a valve between two fluids at high pressure. The valve is typically closed when not in use. The valve opens and allows for the passage of water when a 12 volt DC supply is provided to the two terminals. A 12 volt DC source is required for the solenoid coil, which electronically powers the valve. Since it is a usually closed assembly, the fluid flow begins to open as soon as the power is turned on and stops or is blocked when the supply voltage is cut [18].

IV. PROJECT DESIGN AND WORKING PROCEDURE

The proposed system's schematic circuit design is shown in Fig. 3. Here we conceived and developed the complete concept into a gadget that combines all of the previously described components into a single unit. To create a unique saline supply line, we had to incorporate a solenoid valve in the middle of the supply line, allowing the supply line to be opened or stopped at any moment. The saline bottle is linked to one terminal of the solenoid valve, and the IV drip set is attached to the other. We begin by connecting the load cell to the Arduino using the HX711 module. The HX711 module is useful for detecting loads imposed on the load cell.

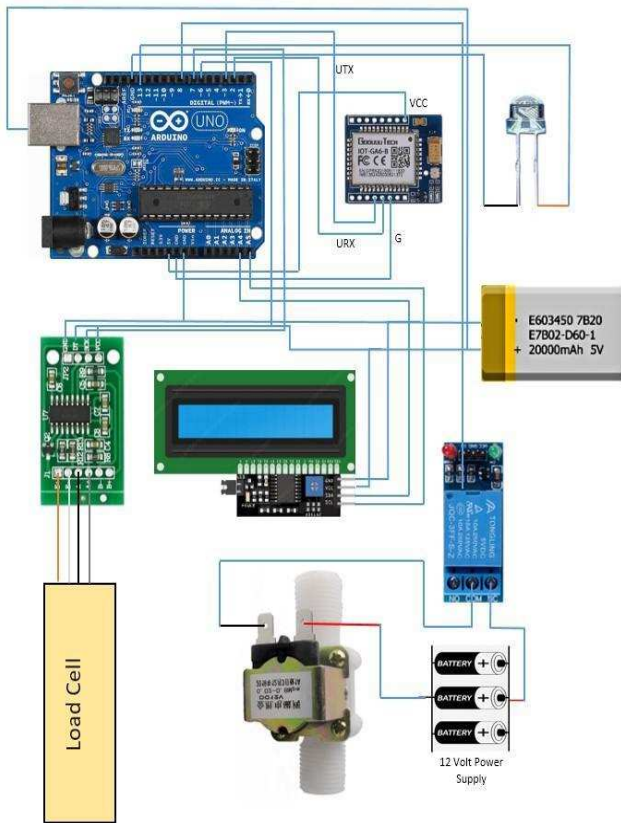


Fig. 3. Circuit diagram of the proposed IV fluid monitoring system

A hook is used to suspend the saline bottle from the load cell. The weight of the saline bottle is detected by the load cell. In the beginning, we established some threshold values (7%,

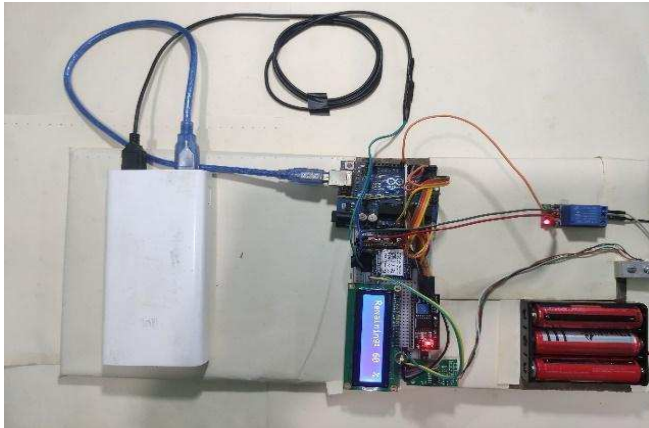
10%, and 15% of the original value of the saline bag) in the code. The load cell is linked to HX711 as follows:

E+ (Excitation+)	➔	RED (V_{cc})
E- (Excitation-)	➔	BLACK (GND)
A- (Amplifier-)	➔	WHITE
A+ (Amplifier+)	➔	GREEN

HX711 is linked to Arduino Uno through V_{cc} to 5V, GND of HX711 to Arduino GND, SCK of HX711 to Arduino D6 (here, D stands for digital pin), and DT of HX711 to Arduino D7. If the saline bottle's weight exceeds the threshold, our system sends an alert notification to hospital administration via the GA6-B GSM module, which operates on an electric signal generated by the Arduino for a specific threshold. GSM GA-6 is connected to the Arduino Uno via V_{cc} of GSM to 5V of Arduino; similarly, GND, UTX, and RTX are assembled with Arduino GND and GPIO pins D3, D2, respectively, and communicate through serial communication to the microcontroller. In the alert notification sent, we can see the current weight percentage of the saline bottle. If the nurse does not quickly respond to the patient, a line blockage arrangement is made utilizing a solenoid valve. This solenoid valve has been opened since the start of the experiment by a 12V battery power supply. A solenoid valve is a valve that is operated by electricity. A solenoid, which is an electric coil with a moveable ferromagnetic core (plunger) in its middle, is used in the valve. In the rest position, the plunger closes a small aperture. A magnetic field is created by passing an electric current through the coil. The magnetic field pulls the plunger higher, opening the aperture. When the saline bag weight reaches 7% of its original value, Arduino cuts off the power supply of the solenoid valve via an electric signal through a relay, which controls the 12 V battery power supply, resulting in closing the solenoid valve. So the saline tube is blocked. This prevents any chance of saline and blood flowing upward from the veins to the bottle.

The single-channel relay is connected to the Arduino Uno through IN to D8, V_{cc} to 5V, and the GND of the relay to the Arduino GND. The 12V battery power supply and solenoid valve are connected in series. Now Relay is connected in parallel with them. The NC terminal of the relay is connected to the 12V battery power supply's positive (+ve) terminal. The negative (-ve) terminal of the battery power supply is connected to one pin of the solenoid valve. Another pin of the solenoid valve is connected to the COM of the relay. The connection between the valve, saline bottle, and IV drip set is fully water-sealed to prevent fluid loss. In addition, one LED light is added to the circuit and linked to the Arduino. One terminal of the LED is linked to the Arduino's D13 pin, while the other terminal is connected to the Arduino's GND pin. An LCD shows the saline bag weight percentage in real time so that the bag weight can be monitored at any time. This LCD is placed locally with the rest of the circuit, so anyone near the patient can monitor the saline bag weight. When the weight percentage drops below 15%, an LED light that serves as an alarm light illuminates. This LED light is installed in

the nursing station. This form of indicator has the advantage of not making any noise, which might bother the patient, especially at night. Every pin on the 16×2 display is connected to the I2C bus, meaning they work as a single device. Now the I2C is connected to the Arduino Uno through SCL to A5 (here A stands for analogue), SDA of I2C to A4, Vcc to 5V, and GND of I2C to Arduino GND.

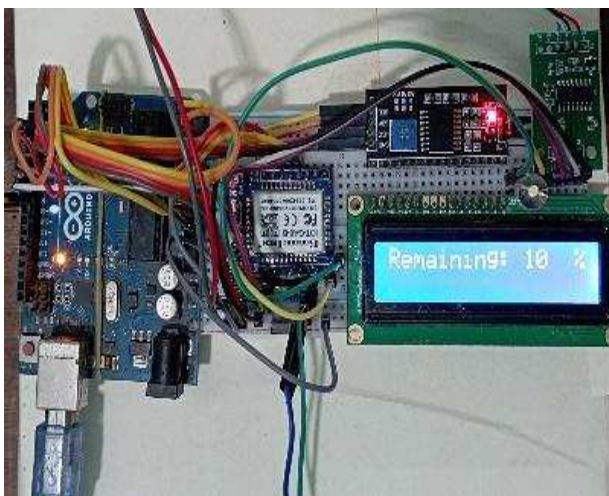


(a) Top view

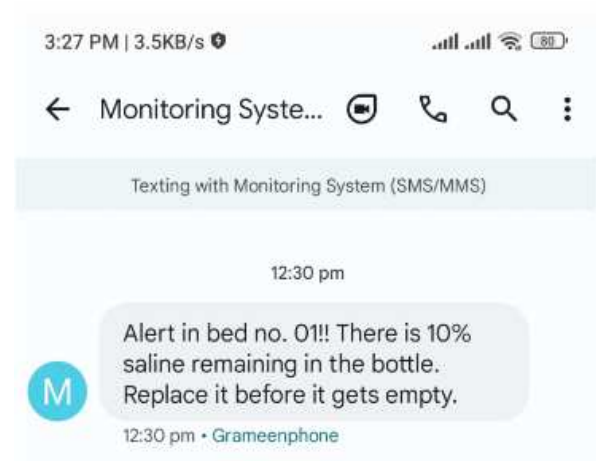


(b) Side view

Fig. 4. Prototype of the proposed IV fluid monitoring system



(a) Remaining percentage



(b) SMS alert

Fig. 5. Generated notification at 10% saline.

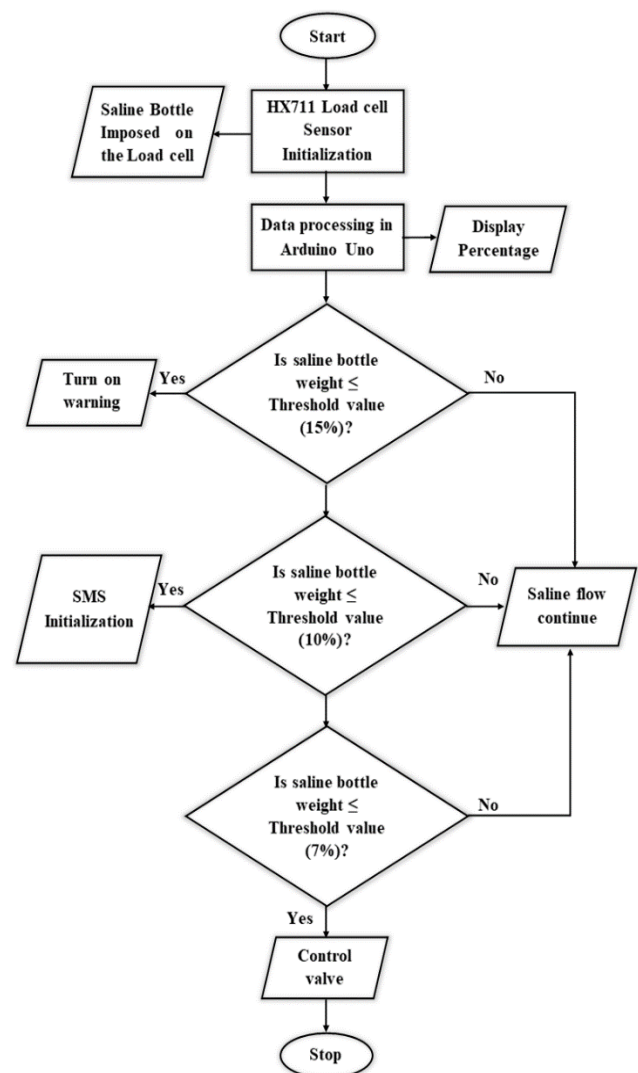


Fig. 6. Flow chart of the of the proposed IV fluid monitoring system

The entire proposed system is demonstrated in Fig. 4 (a) and 4(b) by real-time testing. The programming is done on the Arduino platform with the C language. The program is a collection of instructions that instruct the board on what

duties they must carry out. The developed program adds the parameters for which the valve will shut down. The findings include the proportion of saline weight remaining in the saline bottle, as shown in Fig. 5(a). An SMS alert is sent via the mobile network to the duty nurse or doctor when the remaining saline reaches 10%, as shown in Fig. 5(b). When the remaining percentage reaches 7%, the solenoid valve closes immediately to remove the possibility of reverse blood flow. Fig. 6 shows a flowchart that illustrates the signal flow of the whole proposed IV fluid monitoring system and the decision-making scenario.

V. CONCLUSION

Traditionally, the doctor or nurse manually checks and regulates the saline flow rate based on the patient's demands. However, in cases where nurses are few, the developed prototype can play an important role in monitoring the intravenous (IV) bag of a patient, and this will contribute to the development of a smart healthcare system. In this paper, we have addressed a smart fluid monitoring system where we have basically tried to solve real-life problems with the automated and cost-effective system that people can directly apply in their real life. This technology is especially useful at night since it eliminates the need for humans to continuously monitor the fluid level. The essential part of our system is its hardware setup, including auto shutdown of the valve to prevent reverse IV flow, as well as software programming and analysis of sensor data to convert the system into a GSM-based automatic monitoring system. It can be simply deployed in just about any healthcare setting and allows nurses to quickly observe the fluid flow and securely control it. There is no doubt about connectivity because GSM technology is more reliable than any other techniques mentioned in the literature. Therefore, the proposed system overcomes previous drawbacks and creates a more user-friendly system.

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